AR-Based Intelligent Automobile Repair Learning & Evaluation System.

R26-IT-049

Project Proposal Report

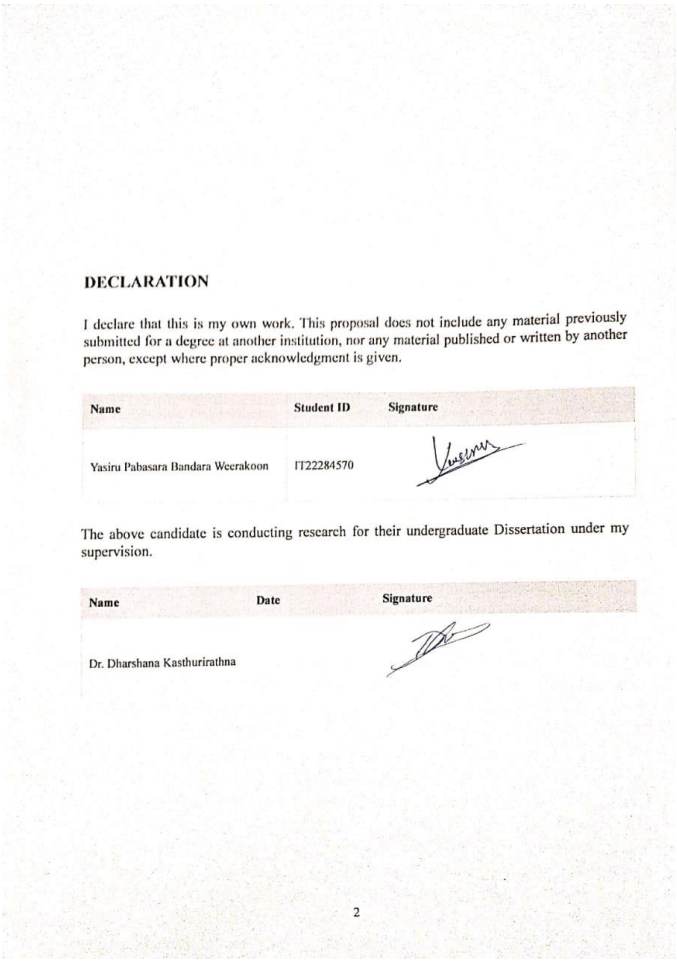
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March 2026

DECLARATION



DECLARATION

I declare that this is my own work. This proposal does not include any material previously submitted for a degree at another institution, nor any material published or written by another person, except where proper acknowledgment is given.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

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ABSTRACT

Practical knowledge of engine parts and repair techniques is necessary for automotive repair courses. Nonetheless, a lot of students at educational institutions don't have constant guidance from knowledgeable teachers during practical sessions. Because they don't get prompt feedback, pupils frequently do repair assignments poorly. It is impossible to confirm whether a learner finishes a repair operation correctly using traditional training methods like manuals and demonstration films.

An important part of the AR-Based Intelligent Automobiles Repair Learning and Evaluation System is the AR-Supported Task Verification and Instruction System proposed in this study. Using computer vision and machine learning approaches, the suggested module compares students' actions with reference repair processes to determine whether they are performing automotive repair tasks correctly.

The system records video in real time from the student's device and extracts visual elements from the reference repair process as well as the student's actions. The system finds improper actions and missing repair steps by examining tool orientation, component location, and action sequences. The student is then guided by augmented reality overlays that highlight the appropriate parts, show arrows, and provide detailed repair instructions.

The system also offers optional voice guidance, which offers spoken directions and safety alerts if improper operations are recognized. For every repair task, the system produces a task correctness score, error report, and final performance feedback. By offering real-time coaching, automated task verification, and intelligent learning support, the suggested module seeks to enhance automotive repair training.

ACKNOWLEDGEMENT

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AR	Augmented Reality
AI	Artificial Intelligence
CV	Computer Vision
ML	Machine Learning
CNN	Convolutional Neural Network
AR Core	Augmented Reality Platform developed by Google
ARKit	Augmented Reality Framework developed by Apple
SDK	Software Development Kit
UI	User Interface
UX	User Experience
API	Application Programming Interface
FPS	Frames Per Second
IoT	Internet of Things

INTRODUCTION

1.1 Background and Context

Strong practical understanding of engine parts and repair techniques is necessary for automotive maintenance training. To participate in automotive training programs, students must be able to recognize engine components, utilize tools correctly, and adhere to acceptable repair procedures.

However, offering practical automotive instruction presents several difficulties for many training facilities. These include inadequate teacher monitoring, restricted vehicle access, and safety issues during training sessions. Because of this, students frequently find it difficult to complete repair assignments effectively and may start using improper repair techniques.

Recent developments in computer vision (CV), artificial intelligence (AI), and augmented reality (AR) offer new ways to improve real-world learning settings. With the use of augmented reality (AR) technology, students can directly get visual instructions while completing activities by superimposing digital information on real-world items.

The proposed study focuses on creating an AR-Supported Task Verification and Instruction System that employs augmented reality overlays and optional audio instructions to analyze student activities in real time, identify mistakes, and direct students through proper repair methods.

By increasing access to cutting-edge practical learning resources in technical education, our research supports Sustainable Development Goal 4 (Quality Education).

1.2 Literature Review

The application of augmented reality technologies in educational and technical training settings has been the subject of numerous studies. By allowing students to see digital instructions directly on tangible things, augmented reality (AR)-based learning systems increase student engagement and retention.

AR-based solutions can successfully direct users throughout assembly jobs and equipment maintenance operations, according to prior study. These technologies aid users in comprehending intricate mechanical processes by offering visual instructions.

Additionally, object recognition and activity analysis have made extensive use of computer vision algorithms. Convolutional Neural Networks (CNN) are frequently used for camera image analysis and mechanical component detection. These methods enable machines to recognize objects and monitor how tools and components interact.

Human behavior in videos has also been examined using machine learning techniques. Algorithms for action recognition can recognize user movement sequences and categorize them into distinct activities.
Despite these developments, most AR-based training systems now in use mostly concentrate on visualization and do not offer automated user action verification. Many systems are unable to match student behavior with reference protocols and offer insightful feedback when errors are made.
Consequently, an integrated system that incorporates automatic performance evaluation, AR based assistance, and visual task verification is required.

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1.3 Research Gap

While augmented reality technologies have been employed in training applications, existing solutions still have several drawbacks. Most AR training systems offer visual guidance, but they don't check if the user completes the task correctly.
Furthermore, during repair procedures, current systems are unable to identify wrong acts or missing phases. Students could continue to complete assignments improperly in the absence of automated feedback. The lack of integrated solutions that merge real-time AR guidance and performance evaluation with computer vision-based task verification is another drawback. By creating an intelligent AR-supported job verification system that can identify mistakes and offer real-time guidance during car repair training, our research attempts to overcome these constraints.

OBJECTIVE

2.1 Main Objective

To provide an AR-Supported Task Verification and Instruction System that can provide real time augmented reality guidance, automated task evaluation, and a comparison between student repair actions and reference processes for automobile learners.

2.2 Specific Objectives

1. To use computer vision techniques to record and analyze student repair operations in real time.
2. To extract visual elements for similarity analysis from students and reference repair videos.
3. To identify improper component placement, missing stages, and improper tool usage.
4. To offer augmented reality overlays that instruct pupils on proper repair techniques.
5. To introduce optional audio instructions that offer safety alerts and step-by-step voice advice.
6. To produce improvement recommendations and task completion scores based on the identified actions.

METHODOLOGY

The suggested system employs a Design Science Research methodology in which a software artefact is created and assessed to solve a practical training issue.

The method uses a mobile device's camera to record video in real time. Key visual characteristics are extracted from the student's actions using computer vision algorithms. These characteristics are contrasted with system-stored reference repair methods.

To ascertain whether the learner completes the repair work correctly, machine learning-based visual similarity analysis is employed. Corrective instructions are generated by the system when it detects wrong actions.

To assist the student, the augmented reality module shows visual overlays on the camera feed, including arrows, labels, and highlighted regions. Additionally, voice-based assistance and safety warnings are provided by optional audio instructions.

Several technologies will be used to implement the system, including:

- OpenCV for processing computer vision
- PyTorch or TensorFlow for machine learning models
- Unity for creating AR interfaces
- ARKit or AR Core for visualizing augmented reality
- Metrics including job completion time, error detection rate, and task verification accuracy will be used to assess the system.

Tools and Technologies

Tool	Purpose	Why Selected
Unity	Development of the augmented reality mobile application interface	Unity provides strong support for AR development and integrates easily with AR Core and ARKit for cross-platform mobile applications
AR Core / ARKit	Implement augmented reality overlays such as arrows, labels, and highlighted regions	These frameworks provide stable real-time environment tracking and AR visualization for Android and iOS devices
OpenCV	Perform computer vision tasks such as image processing and feature extraction	Perform computer vision tasks such as image processing and feature extraction
TensorFlow / PyTorch	Train and implement machine learning models for visual similarity analysis and task verification	These frameworks provide powerful deep learning tools and strong community support for developing AI models
Mobile Device Camera	Capture real-time student repair actions	Smartphone cameras enable real-time data capture required for visual verification and AR interaction
Python	Develop machine learning algorithms and backend processing modules	Python is widely used in AI and machine learning development due to its simplicity and extensive libraries

Audio Processing Libraries	Analyze student audio input and provide voice guidance	These libraries support audio classification and enable the system to provide optional voice based instructions
Firebase / Cloud Storage	Store reference repair videos, datasets, and evaluation results	Cloud storage ensures secure and scalable data management for training datasets and system outputs

Data Collection and Ethical Considerations

To train and assess the task verification and instruction models, the suggested system needs visual and auditory data. Images, videos, and optional audio recordings of auto repair procedures will be used to gather data. Students will be seen using tools, handling components, and aligning parts during these recordings.

Validation and Performance Metrics

Metric	How Measured	Target
Task Verification Accuracy	Compare system predictions with instructor evaluation of whether the repair task was performed correctly	≥ 85% accuracy
Mistake Detection Rate	Measure how many incorrect actions (wrong tool usage, missing steps, incorrect component placement) are correctly identified by the system	≥ 80% detection rate
AR Guidance Effectiveness	Compare number of correctly completed tasks with and without AR guidance during user testing	At least 20% improvement in task accuracy
Task Completion Time	Measure the time taken by students to complete repair tasks with system assistance	Reduce completion time by 15–20%
System Response Time	Measure the time taken for the system to process camera input and generate feedback	≤ 2 seconds

User Satisfaction Score	Collect user feedback using questionnaires after system usage	≥ 80% positive user feedback

HIGH-LEVEL SYSTEM ARCHITECTURE

AR-Supported Task Verification and Instruction System

The AR-Supported operation Verification and Instruction System oversee using augmented reality overlays to provide real-time instruction and confirm whether a student completes an automotive repair operation correctly. Following the system's identification of the engine component and repair task from earlier system modules, this module begins to function.
A real-time camera feeds from the user's mobile device, student photos or videos, reference repair videos, component identification (Component ID), and task identification (Task ID) are among the inputs that the module gets. The students' activities are analyzed and compared with the reference repair procedure using computer vision and machine learning techniques.

To perform visual verification, the system first extracts key visual elements from the reference video and the student's repair procedure. These characteristics include movement patterns during the repair process, component position, and tool orientation. These characteristics are examined by machine learning algorithms to ascertain whether the learner follows the proper repair procedures.

The system detects inaccurate activities, such as missing steps, wrong tool usage, or poor component alignment, once the visual comparison is finished. The augmented reality module creates visual instruction overlays that show up on the mobile device camera feed based on these findings. These overlays point the student in the direction of the proper restoration technique with arrows, highlighted areas, and labels.

Additionally, the system offers optional audio instructions that offer safety alerts when improper activities are recognized as well as step-by-step voice coaching. By helping students who prefer aural instructions, this feature improves the learning process.

Lastly, the decision logic module creates a task correctness score and performance evaluation report by combining the outcomes of visual verification, AR guiding feedback, and optional audio analysis. To assist the student in completing the repair assignment successfully on subsequent attempts, the system provides a list of errors found, AR visual guiding feedback, and ideas for improvement.

The system's high-level architecture increases the efficacy of car repair training by enabling real-time verification, intelligent guiding, and automated performance evaluation.



Figure 1: Overall System Architecture of the AR-Based Intelligent Automobile

USER REQUIREMENTS

Stakeholder

- Automobile engineering students
- Vocational training institutes
- Automobile instructors
- Automotive training centers

Functional Requirements

FR#	Functional Requirement
FR1	While the learner is performing repairs, the system will record their footage in real time.
FR2	Students' actions will be compared to reference repair processes by the system.
FR3	Incorrect actions and missing repair steps will be detected by the system.
FR4	The camera stream will have AR guidance overlays displayed by the system.
FR5	For task guidance, the system will offer optional audio instructions.
FR6	Task evaluation reports will be produced by the system.

Non-Functional Requirements

Category	Requirement
Performance	The term "real- video input processing" is required.
Accuracy	The task verification accuracy needs to be higher than 85%.
Scalability	Several repair tasks should be supported by the system.
Usability	The user interface ought to be straightforward and simple to comprehend.
Security	Student information needs to be kept safe.

COMMERCIALIZATION PLAN

Technical education institutions, vocational training facilities, and auto training facilities that offer instruction in auto repair and maintenance comprise the major target market for the proposed system. Effective learning resources that enhance hands-on instruction and lessen the requirement for ongoing instructor monitoring are necessary for these schools.

Automobile manufacturing firms that offer training programs for technicians and apprentices, as well as automotive service training organizations, represent another possible market niche.

By increasing the efficacy and efficiency of vehicle repair instruction, the suggested AR Supported Task Verification and Instruction System adds substantial value. The device helps students learn difficult repair methods more efficiently by providing automated performance evaluation, augmented reality instruction, and real-time task verification.
The technology offers interactive visual instructions and instant feedback when students do tasks wrong, in contrast to conventional training methods like manuals and static video lessons. This improves learning results, lowers training errors, and increases safety when doing repairs.
The integration of computer vision, machine learning, and augmented reality technologies allows the system to create an intelligent and immersive learning environment for technical education.

Revenue Model and Pricing Strategy

- The system can be commercialized through multiple revenue models:
- **1. Software Licensing Model**
- Educational institutions and training centers can purchase annual licenses to use the software platform for training purposes.
- **2. Subscription-Based Model**
- Training institutes can subscribe to the platform on a monthly or yearly basis to access the AR learning modules, task verification features, and system updates.
- **3. Enterprise Training Solutions**
- Automobile companies and training organizations can purchase customized versions of the system for technician training programs.

Competitive Advantage



Figure 2: Architecture of the AR-Supported Task Verification and Instruction System

According to cost estimation, most of the development tools needed for the suggested system are open-source or publicly accessible, greatly lowering the total cost of development. For research and academic development, technologies like OpenCV, Python, TensorFlow/PyTorch, and Unity with AR Core/ARKit support are free to use. Because of this, the system's initial build cost is still quite modest. Cloud storage and backend hosting for datasets, reference repair videos, and system updates, as well as possible maintenance and technical assistance for universities utilizing the platform, will be the main continuing expenses.

Key technical advances of this research include the suggested AR-supported task verification framework, the intelligent AR instruction overlay system, and the visual similarity matching method. Furthermore, the task evaluation and automatic feedback generating engine presents a new method for employing computer vision and machine learning approaches to validate realistic car repair operations. These elements may be protected by software copyright or trade secret protection, and if the system is further developed for commercialization, they may be taken into consideration for software patent registration under SLIIT's undergraduate intellectual property policy.

BUDGET AND JUSTIFICATION

Category	Item	Cost (LKR)	Justification
Software	Unity, OpenCV, Python, TensorFlow/PyTorch, Git (all opensource)	0	Open-source development tools used for AR, computer vision, and machine learning
Hardware	1 Android smartphone for AR testing	35,000	Required to test ARCore features and real-time camera processing
Hardware	Tripod / phone holder for recording reference repair videos	3,000	Ensures stable recording for dataset collection
Cloud	Google Colab Pro (1 month) for ML model training/Cloud storage / Firebase for dataset storage	5,000	Stores reference repair videos and system evaluation data
Other	Mobile data for development and testing (12 months)	12,000	Hard copies required for academic submission
Contingency	Reserve for unexpected hardware issues	3,000	10% contingency for hardware-dependent project
TOTAL		**58,000**	

The project budget is kept relatively low because most development tools used, such as Python, OpenCV, TensorFlow/PyTorch, Unity, and AR frameworks, are free and open source. The main cost, totaling approximately **LKR 58,000**, is allocated for hardware devices required to test augmented reality features and real-time computer vision processing on physical mobile devices. Additional minor expenses include cloud services for machine learning experiments, mobile data for development activities, and printing costs for documentation.

WORK BREAKDOWN STRUCTURE (WBS)

The table below shows all tasks broken down by phase across the 18-week semester. Each task has a name deliverable, and the workload is distributed to match the available time per phase.

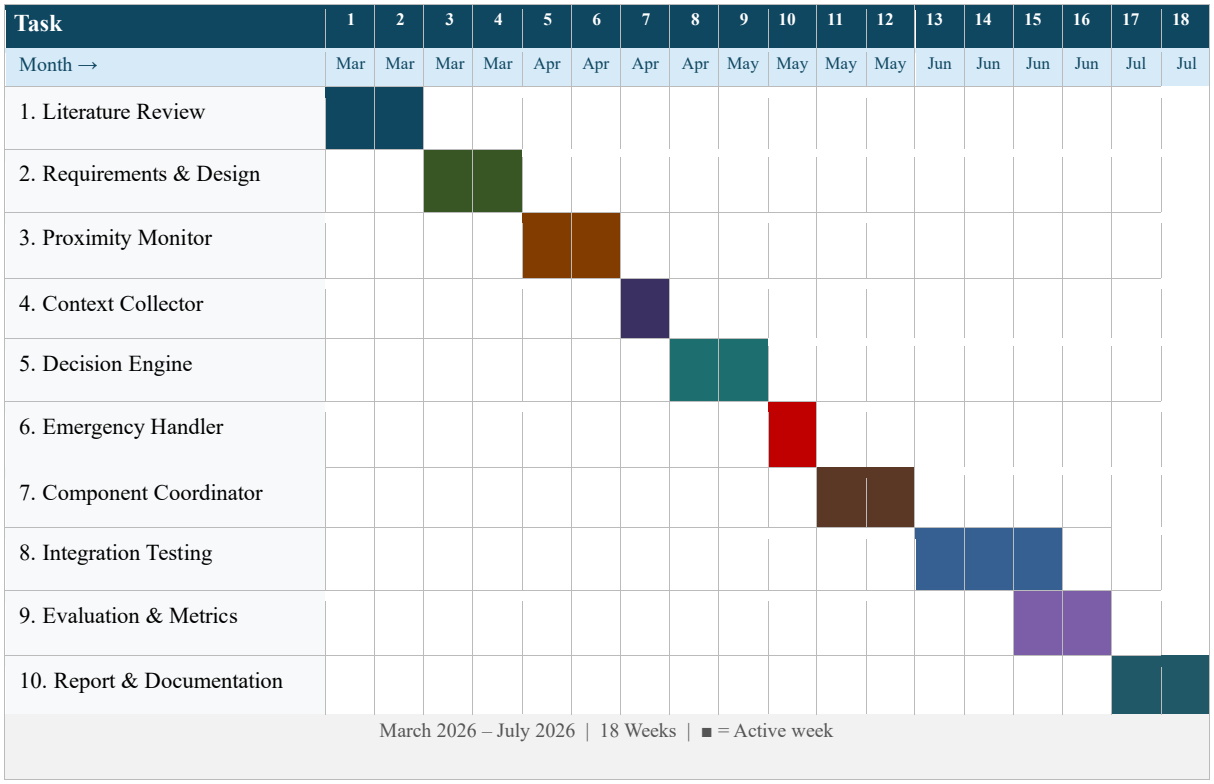
Phase	Task / Activity	Deliverable
Phase 1 Planning Wk. 1–2	Literature review on AR learning systems, computer vision, and task verification methods	Annotated bibliography and research gap analysis
	Define system requirements (FR/NFR) and module scope with team members	Requirements specification document
	Design system workflow for AR supported task verification	System workflow diagram
Phase 2 Design Wk. 3–4	Design visual verification architecture and feature extraction process	System architecture diagram
	Design AR instruction overlay interface and user interaction flow	Design AR instruction overlay interface and user interaction flow
	Design dataset structure for reference repair videos and student recordings	Dataset schema
Phase 3 Dev Wk. 5–12	Implement computer vision module for feature extraction and comparison	Working visual verification module

	Implement AR instruction overlay (arrows, labels, highlighted regions)	AR guidance prototype
	Implement optional audio guidance module	Audio instruction module
	▸ Implement Emergency Handler (flood broadcast + ACK timer)	Emergency delivery < 2 sec
	Develop decision logic for task verification and scoring	Task evaluation algorithm
Phase 4 Testing Wk. 13–16	Conduct unit testing for each system module	Unit test results
	Perform integration testing of full system	Integration test report
	Evaluate system performance (accuracy, response time, usability)	Performance evaluation report
Phase 5 Docs Wk. 17–18	Prepare research proposal and technical documentation	Completed research proposal
	Final review and submission	Submitted proposal

Figure 3: Visual Task Verification Process Flow

GANTT CHART

The chart below shows the full 18-week project timeline from March 2026 to July 2026, with colored bars showing active development weeks and milestone alignment with semester checkpoints.



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APPENDICES

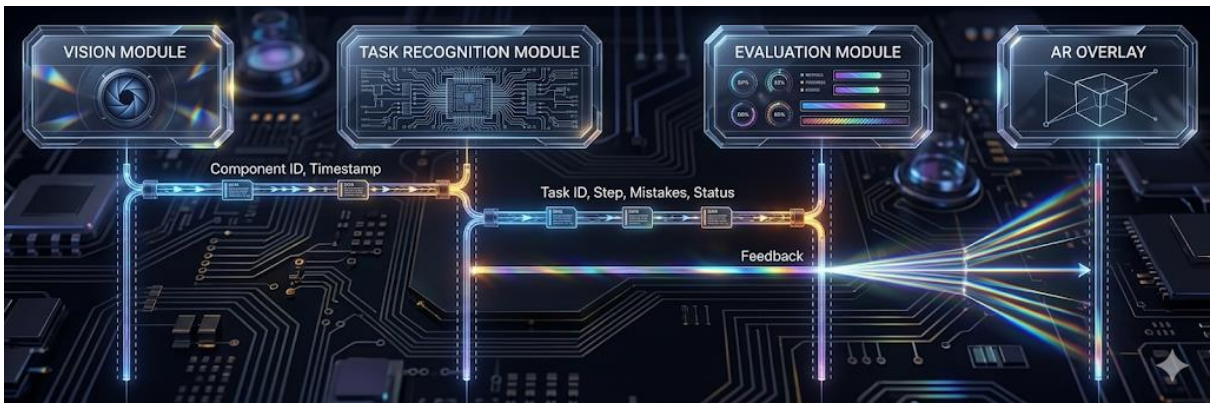


Figure 4: Augmented Reality Instruction Overlay Interface

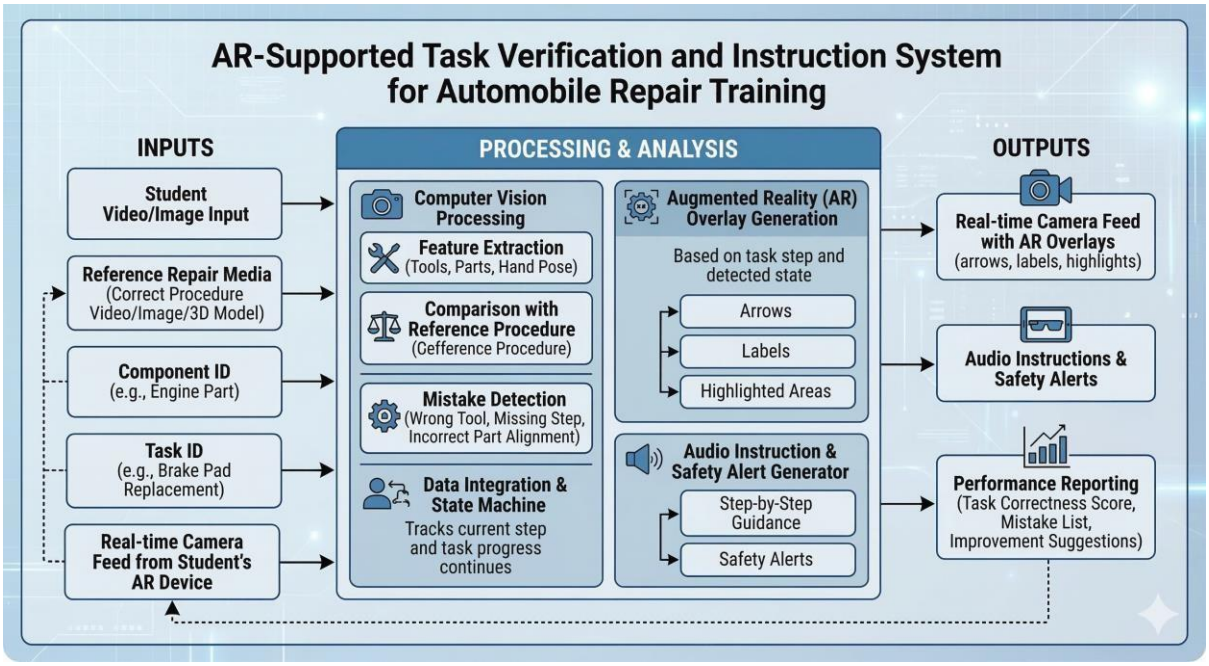


Figure 5: Task Evaluation and Feedback Generation Process